Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Crane Dairy LLC

Physical address of dairy:

Number and Street	City	County	Zip Code
20633 W Crane AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1967

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X230-X039-XXXX

B. OPERATORS

Souza, Rosemary

Operator name: Souza, Rosemary

Telephone no.: (209) 667-4868 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6736 Columbus AVE	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Souza, Rosemary

Legal owner name: Souza, Rosemary

Telephone no.: (209) 667-4868 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6736 Columbus AVE	Hilmar	CA	95324

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	420	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	0	420	0	0
Average number	0	0	0	420	0	0
Avg live weight (lbs)	0	0	0	600		

Predominant milk cow breed: Jersey

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 3,838.61 *tons per reporting period*

Total nitrogen from manure: 39,858.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 27,900.60 *lbs per reporting period*

Total phosphorus from manure: 6,745.20 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: _____ *gallons*

Total nitrogen generated: _____ *lbs*

Total phosphorus generated: _____ *lbs*

Total potassium generated: _____ *lbs*

Total salt generated: _____ *lbs*

	0	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	0	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation Well	Ground water
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	Separator solids *Solid Manure-VDF*	4,224.00 *ton*	Dry-weight	78.0	15,800.00	2,500.00	6,000.00		14.70

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	29,365.25	4,646.40	11,151.36	273,208.32
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	29,365.25	4,646.40	11,151.36	273,208.32

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	Corral solids	145.00 *ton*	Dry-weight	46.2		25,200.00	10,200.00	47,900.00		37.10

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	3,931.70	1,591.40	7,473.36	57,883.42
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	3,931.70	1,591.40	7,473.36	57,883.42

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
B51-B54	59	59	2	manure	X045-X230-X008-XXXX
B55-B56	19	19	2	manure	X045-X230-X031-XXXX
B57	19	19	2	manure	X045-X230-X018-XXXX
B58-B60	35	35	2	manure	X045-X230-X038-XXXX
Totals for areas that were used for application	132	132	8		
Totals for areas that were not used for application					
Land application area totals	132	132	8		

B. CROPS AND HARVESTS

B51-B54

Field name: B51-B54

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 59 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	979.69 *ton*	As-is		67.6	4,800.00	900.00	5,100.00		7.12

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.60	159.41	29.89	169.37	766.11

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 59 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,489.01 *ton*	As-is		71.2	3,800.00	600.00	3,800.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.24	191.80	30.28	191.80	1,482.75

B55-B56

Field name: B55-B56

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	326.89 *ton*	As-is		65.0	4,900.00	1,100.00	6,100.00		6.93

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.20	168.61	37.85	209.90	834.60

06/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	473.67 *ton*	As-is		71.2	3,800.00	600.00	3,800.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.93	189.47	29.92	189.47	1,464.69

B57

Field name: B57

10/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 10/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	223.82 *ton*	As-is		54.0	5,700.00	1,100.00	5,800.00		5.61

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.78	134.29	25.92	136.65	607.99

B57

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/19/2023	391.78 *ton*	As-is		72.2	4,400.00	400.00	4,600.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.62	181.46	16.50	189.70	1,146.47

B58-B60

Field name: B58-B60

11/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	429.27 *ton*	As-is		64.0	4,900.00	1,000.00	5,400.00		6.65

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.26	120.20	24.53	132.46	587.24

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	999.01 *ton*	As-is		68.4	3,700.00	600.00	4,600.00		8.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.54	211.22	34.25	262.60	1,461.18

NUTRIENT BUDGET

A. LAND APPLICATIONS

B51-B54 - 11/12/2022: Oats, silage-soft dough

Field name: B51-B54

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	158.92	5,761,912.00 *gal*
Application event totals		0.00	0.00	0.00	158.92	

B51-B54 - 06/14/2023: Corn, silage

Field name: B51-B54

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	222.46	35.20	84.48	2,069.76	1,888.00 *ton*
Application event totals		222.46	35.20	84.48	2,069.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	343.64	12,459,157.00 *gal*
Application event totals		0.00	0.00	0.00	343.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	293.48	10,640,635.00 *gal*
Application event totals		0.00	0.00	0.00	293.48	

B51-B54 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	238.29	8,639,609.00 *gal*
Application event totals		0.00	0.00	0.00	238.29	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	244.13	8,851,444.00 *gal*
Application event totals		0.00	0.00	0.00	244.13	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	227.59	8,251,788.00 *gal*
Application event totals		0.00	0.00	0.00	227.59	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	173.57	6,293,129.00 *gal*
Application event totals		0.00	0.00	0.00	173.57	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	127.82	4,634,298.00 *gal*
Application event totals		0.00	0.00	0.00	127.82	

09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	116.40	4,220,405.00 *gal*
Application event totals		0.00	0.00	0.00	116.40	

B55-B56 - 11/12/2022: Oats, silage-soft dough

B55-B56 - 11/12/2022: Oats, silage-soft dough

Field name: B55-B56

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	139.00	1,622,982.00 *gal*
Application event totals		0.00	0.00	0.00	139.00	

B55-B56 - 06/21/2023: Corn, silage

Field name: B55-B56

Crop: Corn, silage

Plant date: 06/21/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	222.46	35.20	84.48	2,069.76	608.00 *ton*
Application event totals		222.46	35.20	84.48	2,069.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	222.46	2,597,423.00 *gal*
Application event totals		0.00	0.00	0.00	222.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	222.46	2,597,423.00 *gal*
Application event totals		0.00	0.00	0.00	222.46	

B55-B56 - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	214.10	2,499,825.00 *gal*
Application event totals		0.00	0.00	0.00	214.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	247.30	2,887,474.00 *gal*
Application event totals		0.00	0.00	0.00	247.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	185.34	2,163,976.00 *gal*
Application event totals		0.00	0.00	0.00	185.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	159.38	1,860,889.00 *gal*
Application event totals		0.00	0.00	0.00	159.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	159.38	1,860,889.00 *gal*
Application event totals		0.00	0.00	0.00	159.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	173.83	2,029,672.00 *gal*
Application event totals		0.00	0.00	0.00	173.83	

B57 - 10/03/2022: Oats, silage-soft dough

B57 - 10/03/2022: Oats, silage-soft dough

Field name: B57

Crop: Oats, silage-soft dough

Plant date: 10/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	166.01	74.38	90.01	2,333.87	95.00 *ton*
Application event totals		166.01	74.38	90.01	2,333.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	241.98	2,825,300.00 *gal*
Application event totals		0.00	0.00	0.00	241.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	416.17	4,859,169.00 *gal*
Application event totals		0.00	0.00	0.00	416.17	

B57 - 06/02/2023: Corn, silage

Field name: B57

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	222.46	35.20	84.48	2,069.76	608.00 *ton*
Application event totals		222.46	35.20	84.48	2,069.76	

B57 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	346.48	4,045,445.00 *gal*
Application event totals		0.00	0.00	0.00	346.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	254.09	2,966,716.00 *gal*
Application event totals		0.00	0.00	0.00	254.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	235.67	2,751,622.00 *gal*
Application event totals		0.00	0.00	0.00	235.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	235.67	2,751,622.00 *gal*
Application event totals		0.00	0.00	0.00	235.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	250.33	2,922,810.00 *gal*
Application event totals		0.00	0.00	0.00	250.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	226.65	2,646,308.00 *gal*
Application event totals		0.00	0.00	0.00	226.65	

B57 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	158.11	1,846,113.00 *gal*
Application event totals		0.00	0.00	0.00	158.11	

B58-B60 - 11/12/2022: Oats, silage-soft dough

Field name: B58-B60

Crop: Oats, silage-soft dough

Plant date: 11/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	201.83	4,340,988.00 *gal*
Application event totals		0.00	0.00	0.00	201.83	

B58-B60 - 06/14/2023: Corn, silage

Field name: B58-B60

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	222.46	35.20	84.48	2,069.76	1,120.00 *ton*
Application event totals		222.46	35.20	84.48	2,069.76	

B58-B60 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	289.56	6,227,949.00 *gal*
Application event totals		0.00	0.00	0.00	289.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	307.29	6,609,252.00 *gal*
Application event totals		0.00	0.00	0.00	307.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	216.98	4,666,888.00 *gal*
Application event totals		0.00	0.00	0.00	216.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	216.98	4,666,888.00 *gal*
Application event totals		0.00	0.00	0.00	216.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	210.77	4,533,269.00 *gal*
Application event totals		0.00	0.00	0.00	210.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	151.37	3,255,741.00 *gal*
Application event totals		0.00	0.00	0.00	151.37	

B58-B60 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	121.01	2,602,734.00 *gal*
Application event totals		0.00	0.00	0.00	121.01	

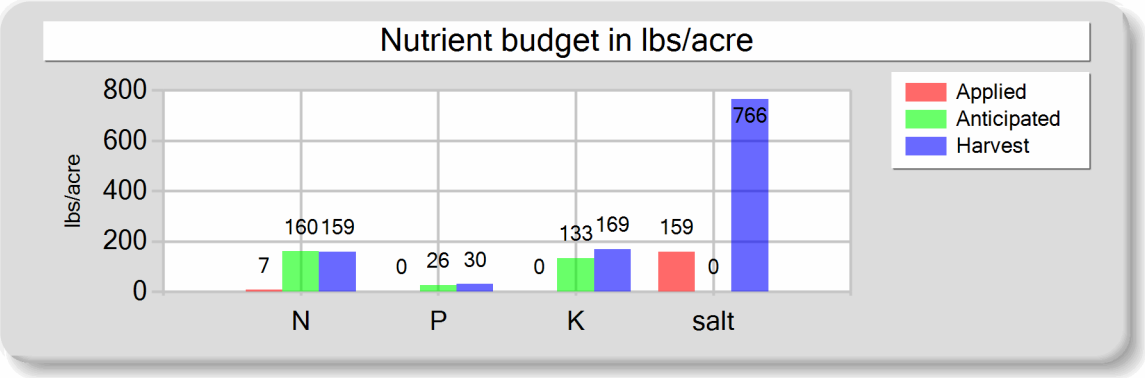
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	192.88	4,148,556.00 *gal*
Application event totals		0.00	0.00	0.00	192.88	

B. NUTRIENT BUDGET

B51-B54 - 11/12/2022: Oats, silage-soft dough

Field name: B51-B54 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	158.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	158.92
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	159.41	29.89	169.37	766.11
Nutrient balance	-152.41	-29.89	-169.37	-607.19
Applied to removed ratio	0.04	0.00	0.00	0.21

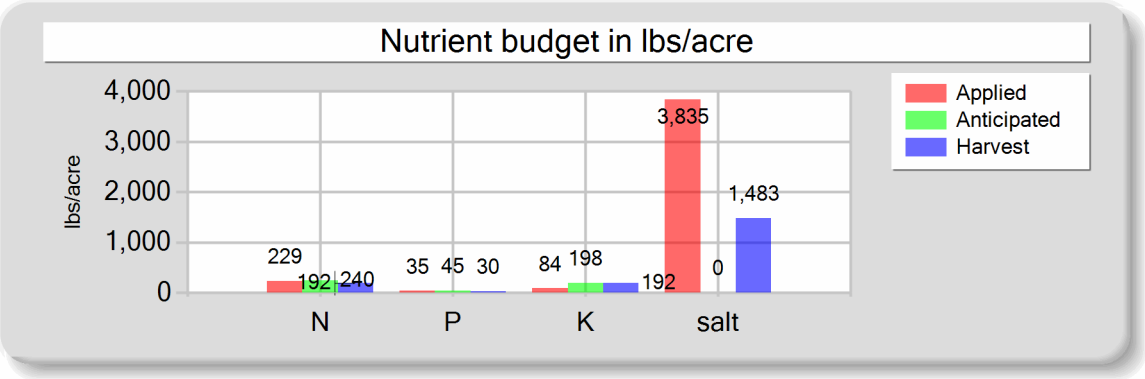
Fresh water applied	
5,761,912.00	*gallons*
212.19	*acre-inches*
3.60	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B51-B54 - 06/14/2023: Corn, silage

Field name: B51-B54 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	222.46	35.20	84.48	2,069.76
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,764.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.46	35.20	84.48	3,834.68
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	191.80	30.28	191.80	1,482.75
Nutrient balance	37.66	4.92	-107.32	2,351.93
Applied to removed ratio	1.20	1.16	0.44	2.59

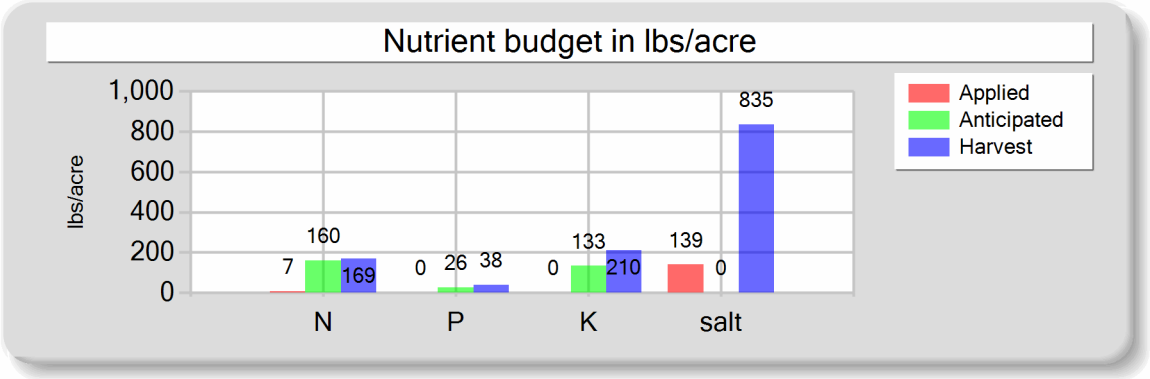
Fresh water applied	
63,990,465.00	*gallons*
2,356.55	*acre-inches*
39.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B55-B56 - 11/12/2022: Oats, silage-soft dough

Field name: B55-B56 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	139.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	139.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	168.61	37.85	209.90	834.60
Nutrient balance	-161.61	-37.85	-209.90	-695.60
Applied to removed ratio	0.04	0.00	0.00	0.17

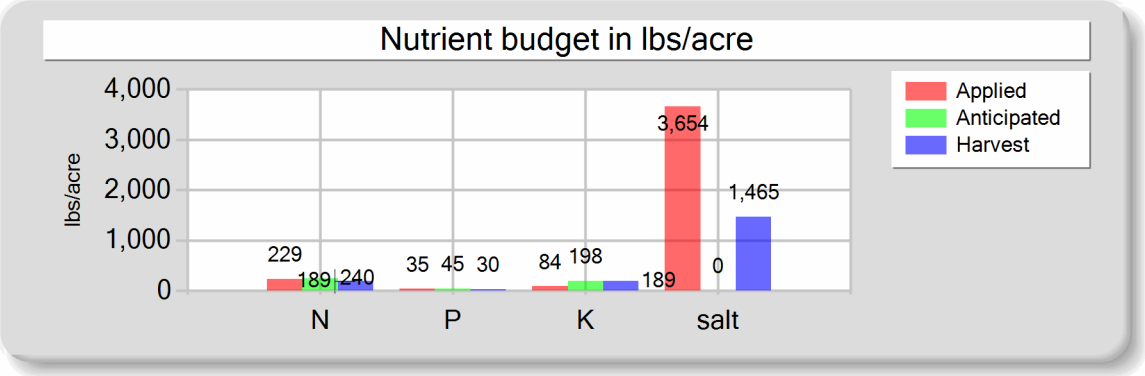
Fresh water applied	
1,622,982.00	*gallons*
59.77	*acre-inches*
3.15	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B55-B56 - 06/21/2023: Corn, silage

Field name: B55-B56 Crop: Corn, silage Plant date: 06/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	222.46	35.20	84.48	2,069.76
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,584.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.46	35.20	84.48	3,654.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	189.47	29.92	189.47	1,464.69
Nutrient balance	40.00	5.28	-104.99	2,189.32
Applied to removed ratio	1.21	1.18	0.45	2.49

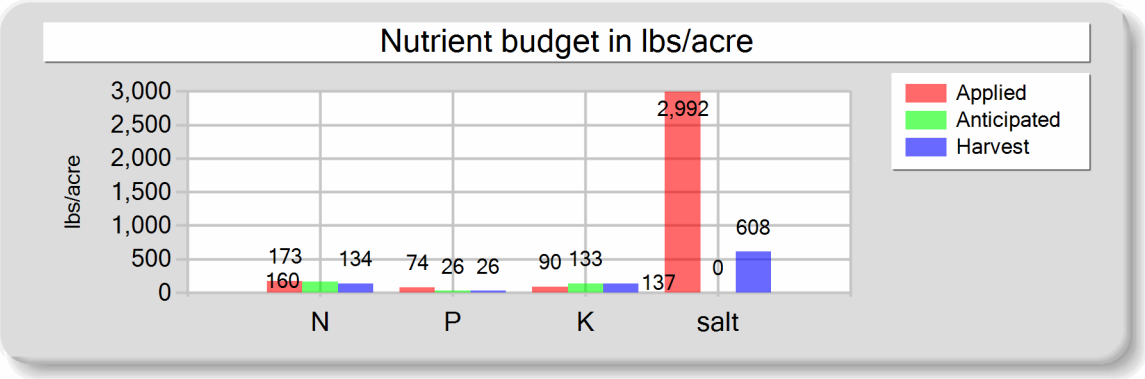
Fresh water applied	
18,497,571.00	*gallons*
681.20	*acre-inches*
35.85	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B57 - 10/03/2022: Oats, silage-soft dough

Field name: B57 Crop: Oats, silage-soft dough Plant date: 10/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	166.01	74.38	90.01	2,333.87
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	658.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.01	74.38	90.01	2,992.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	134.29	25.92	136.65	607.99
Nutrient balance	38.72	48.47	-46.64	2,384.03
Applied to removed ratio	1.29	2.87	0.66	4.92

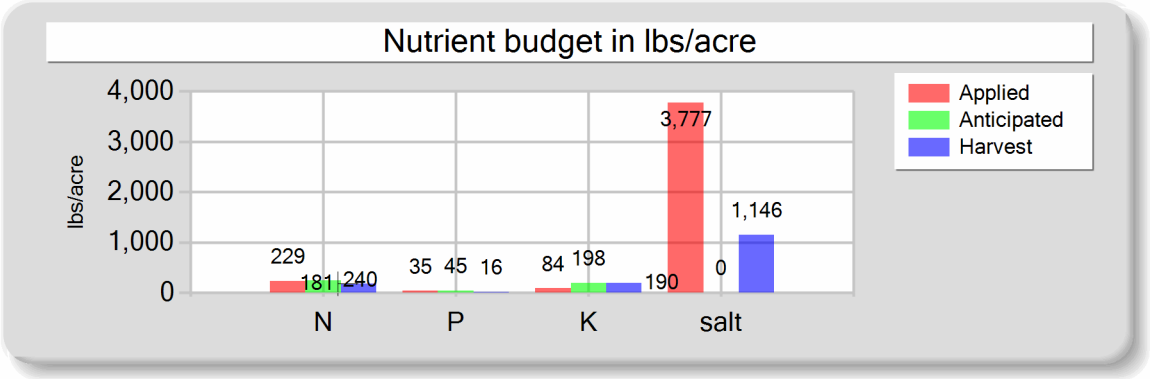
Fresh water applied	
7,684,469.00	*gallons*
282.99	*acre-inches*
14.89	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B57 - 06/02/2023: Corn, silage

Field name: B57 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	222.46	35.20	84.48	2,069.76
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,706.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.46	35.20	84.48	3,776.74
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	181.46	16.50	189.70	1,146.47
Nutrient balance	48.01	18.70	-105.22	2,630.27
Applied to removed ratio	1.26	2.13	0.45	3.29

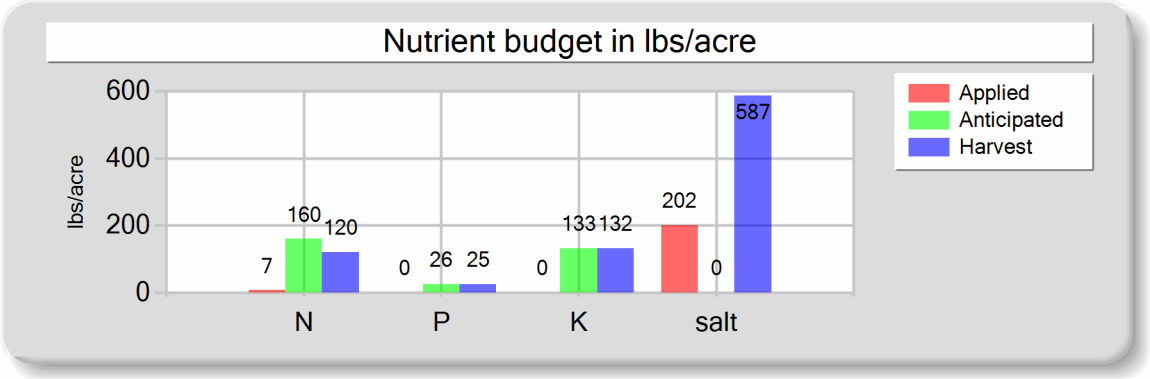
Fresh water applied	
19,930,636.00	*gallons*
733.98	*acre-inches*
38.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B58-B60 - 11/12/2022: Oats, silage-soft dough

Field name: B58-B60 Crop: Oats, silage-soft dough Plant date: 11/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	201.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	7.00	0.00	0.00	201.83
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	120.20	24.53	132.46	587.24
Nutrient balance	-113.20	-24.53	-132.46	-385.41
Applied to removed ratio	0.06	0.00	0.00	0.34

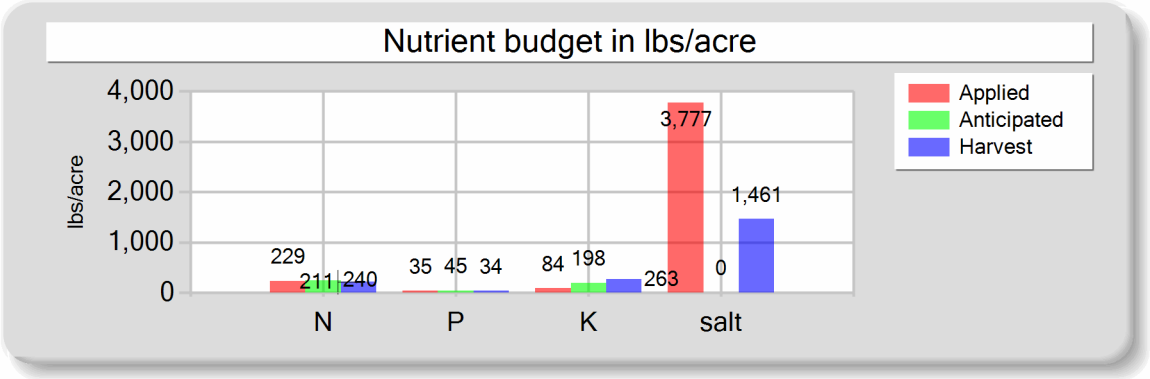
Fresh water applied	
4,340,988.00	*gallons*
159.86	*acre-inches*
4.57	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B58-B60 - 06/14/2023: Corn, silage

Field name: B58-B60 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	222.46	35.20	84.48	2,069.76
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,706.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.46	35.20	84.48	3,776.60
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	211.22	34.25	262.60	1,461.18
Nutrient balance	18.24	0.95	-178.12	2,315.42
Applied to removed ratio	1.09	1.03	0.32	2.58

Fresh water applied	
36,711,277.00	*gallons*
1,351.95	*acre-inches*
38.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Imported Solid Manure Spring VDF

Sample and source description: Imported Solid Manure Spring VDF

Sample date: 05/01/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	2,500.00	6,000.00							14.70
DL	100.00	100.00	100.00							0.01

Spring Solid Manure

Sample and source description: Spring Solid Manure

Sample date: 05/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,600.00	9,800.00	43,400.00							40.50
DL	100.00	100.00	100.00							0.01

Fall Solid Manure

Sample and source description: Fall Solid Manure

Sample date: 12/11/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,200.00	10,200.00	47,900.00	25,500.00	9,900.00	18,000.00	6,400.00	2,490.00		37.10
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

No process wastewater analyses entered.

C. FRESH WATER ANALYSES

Irrigation Well

Ag Well #1

Sample description: Ag Well #1

Sample date: 10/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	16.30	0.00	10.70								875.00	455
DL	0.50	0.50	0.50								1.00	10

T.I.D Canal

Canal

Sample description: Canal

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	195
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

B51-B54 - 11/12/2022: Oats, silage-soft dough

B51-B54 - 11/12/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	900.00	5,100.00		7.12
DL	100.00	100.00	100.00		0.01

B51-B54 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	600.00	3,800.00		10.20
DL	100.00	100.00	100.00		0.01

B55-B56 - 11/12/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	1,100.00	6,100.00		6.93
DL	100.00	100.00	100.00		0.01

B55-B56 - 06/21/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,800.00	600.00	3,800.00		10.20
DL	100.00	100.00	100.00		0.01

B57 - 10/03/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,700.00	1,100.00	5,800.00		5.61
DL	100.00	100.00	100.00		0.01

B57 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	400.00	4,600.00		10.00
DL	100.00	100.00	100.00		0.01

B58-B60 - 11/12/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,900.00	1,000.00	5,400.00		6.65
DL	100.00	100.00	100.00		0.01

B58-B60 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/10/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,700.00	600.00	4,600.00		8.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

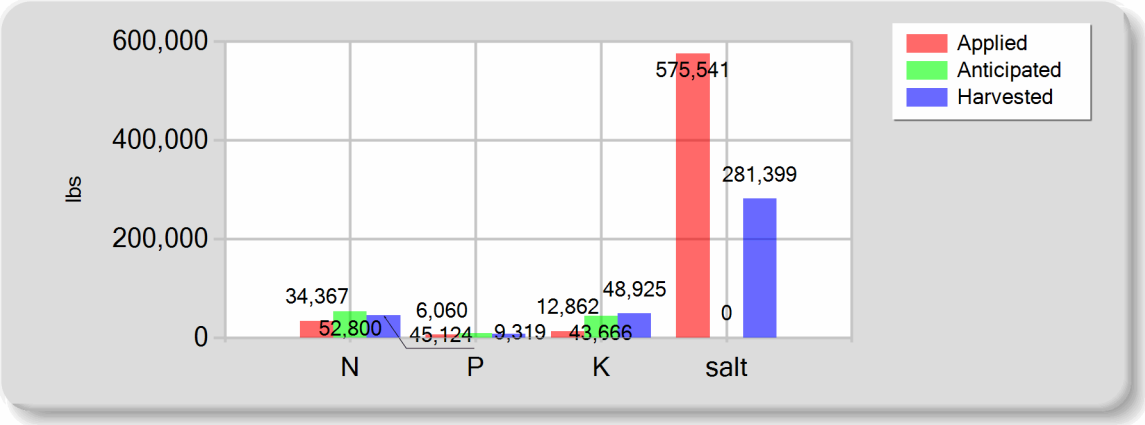
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

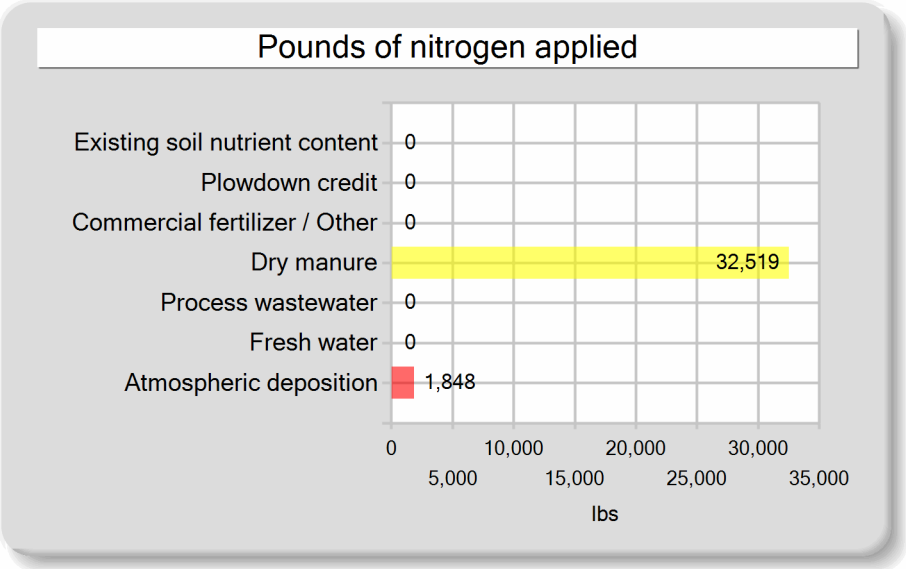
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

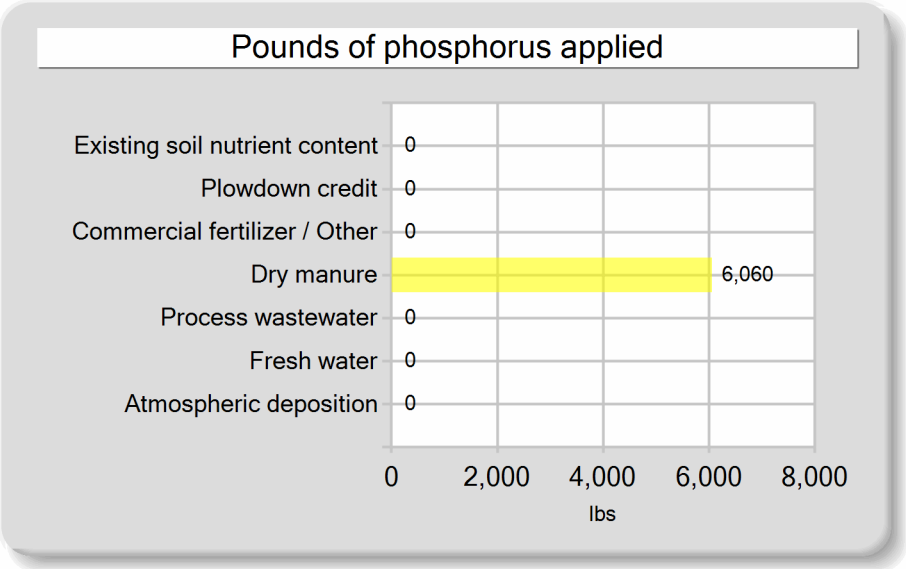
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	32,519.48	6,059.66	12,861.61	317,551.85
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	257,988.67
Atmospheric deposition	1,848.00	0.00	0.00	0.00
Total nutrients applied	34,367.48	6,059.66	12,861.61	575,540.52
Anticipated crop nutrient removal	52,800.00	9,319.20	43,665.60	0.00
Actual crop nutrient removal	45,123.65	7,701.00	48,924.96	281,398.89
Nutrient balance	-10,756.17	-1,641.34	-36,063.35	294,141.63
Applied to removed ratio	0.76	0.79	0.26	2.05

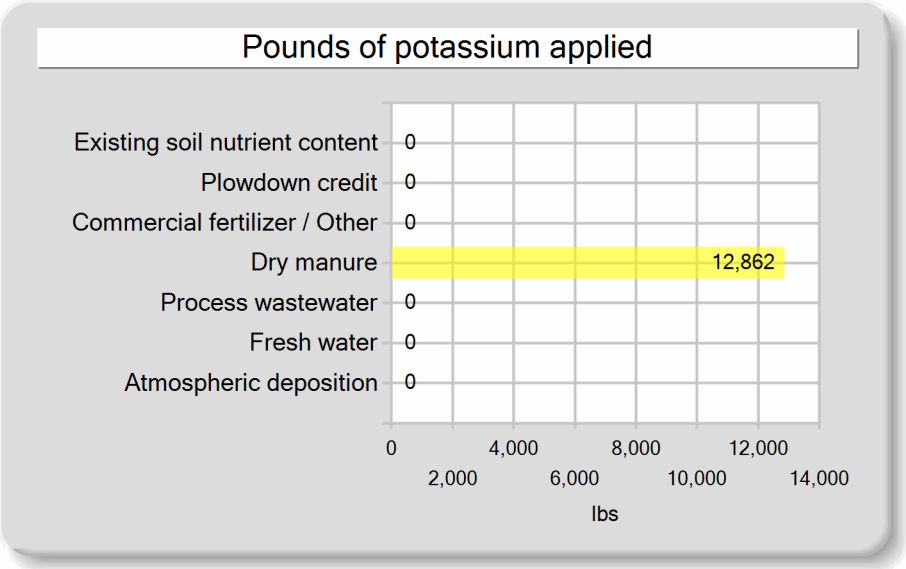
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

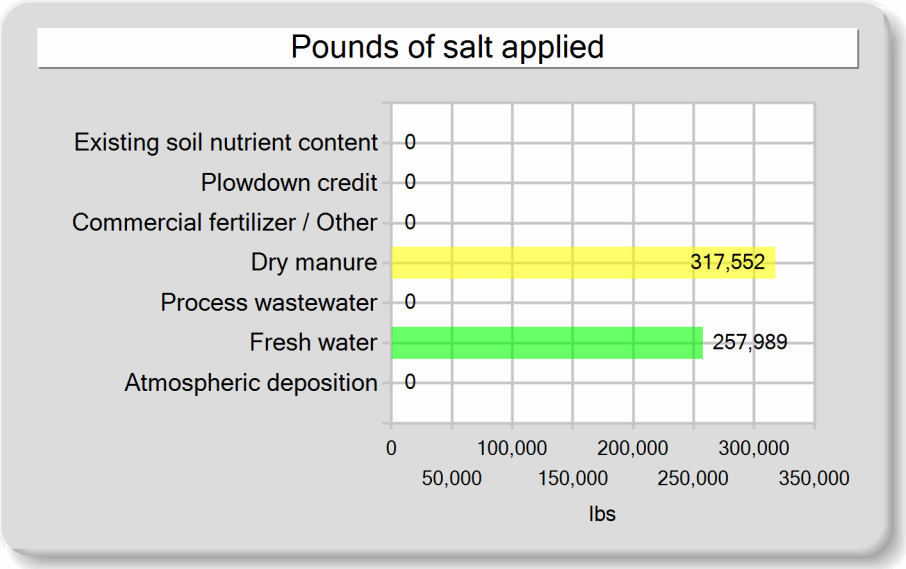


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Facility only housing heifers onsite. Facility is dry scraped, material is either mounded in empty corrals or spread onto land application areas for crops. Fields are being leased out to a third party.

Winter Forage fields B51-B54, B55-B56, and B58-B60 were not able to have any nutrient applications before forage planting due to a late plant date from sudan crop in late 2022.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Rosemary Souza Rosemary Souza (May 20, 2024 13:02 PDT)	Rosemary Souza Rosemary Souza (May 20, 2024 13:02 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Rosemary Souza	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 20, 2024	May 20, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Crane Dairy Inc.
6736 Columbus Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 22.3c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #1	T3303039A	Electrical Conductivity (EC)	EPA 120.1 and	0.879	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	563	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)	1.29
Electrical Conductivity (EC)	100			0.888
Nitrate Nitrogen (NO3-N)	101	99.4 (W3303001A)	94.5 (W3303001A)	5.03

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Crane Dairy Inc.
6736 Columbus Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 20.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #2	T3303039B	Electrical Conductivity (EC)	EPA 120.1 and	0.874	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	559	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)	1.29
Electrical Conductivity (EC)	97.1			0.309
Nitrate Nitrogen (NO3-N)	101	99.4 (W3303001A)	94.5 (W3303001A)	5.03

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Compliance Analysis Report

Customer
Crane Dairy Inc.
6736 Columbus Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #3	T3303039C	Electrical Conductivity (EC)	EPA 120.1 and	0.876	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	561	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)	1.29
Electrical Conductivity (EC)	97.1			0.309
Nitrate Nitrogen (NO3-N)	101	99.4 (W3303001A)	94.5 (W3303001A)	5.03

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Compliance Analysis Report

Customer
Crane Dairy Inc.
6736 Columbus Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #4	T3303039D	Electrical Conductivity (EC)	EPA 120.1 and	0.875	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	560	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)	1.29
Electrical Conductivity (EC)	97.1			0.309
Nitrate Nitrogen (NO3-N)	101	99.4 (W3303001A)	94.5 (W3303001A)	5.03

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

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www.denelelabs.com

Compliance Analysis Report

Customer
Crane Dairy Inc.
6736 Columbus Ave.
Hilmar, CA 95324

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 20.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well #1	T3303039E	Electrical Conductivity (EC)	EPA 120.1 and	0.875	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.7	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	455	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	16.3	mg/L
		Soluble Salts (SALT-SOL)	Calculation	560	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	97.8	93.4 (T3303035A)	92.2 (T3303035A)			1.29
Electrical Conductivity (EC)	97.1			0.874 (T3303039B)	0.876 (Dup)	0.309
Nitrate Nitrogen (NO3-N)	101	99.4 (W3303001A)	94.5 (W3303001A)			5.03
Total Dissolved Solids (TDS)	99.0			500 (T3303041E)	490 (Dup)	2.02
Total Nitrogen (TN)	91.3			2.80 (T3297037A)	2.80 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Crane Dairy
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F & R Ag Services
Sampled By: K.Jones
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	10/30	12:00	Domestic Wells	DW #1	22.3		X				
2		12:15	Domestic Wells	DW #2	20.4		X				
3		12:30	Domestic Wells	DW #3	21.2		X				
4		12:45	Domestic Wells	DW #4	21.0		X				
5		1:00	Irrigation Well	Ag Well #1	20.2						H7
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged In By:	LABORATORY NUMBER
1st		F & R Ag Services				
2nd		DENELE ANALYTICAL, INC.	10/30/23 14:35			T3303039

Required to be submitted to any agency? (YES / NO)

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Rosemary Sousa

Name of Dairy Facility: Crane Dairy

Facility Address: 20033 W. Crane Ave Hilmar 95324
Number and Street City Zip Code

Contact Person Name and Phone Number: Rosemary Sousa (209) 648-1602
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Dores Ag Services

Address of Hauling Company /Person: 22846 Second Ave Stevinson 95324
Number and Street City Zip Code

Contact Person: Phil Dores (209) 649-4503
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Clauss Dairy Farms 21670 Bloss Ave Hilmar 95324 (209) 648-1602
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

645-230-054
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: 10/23/23

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 145 Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): @ 53.8%
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: @ 12 tons per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount:

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Rosemary Souza
Rosemary Souza (May 20, 2024 13:02 PDT) Date: May 20, 2024

Hauler's Signature: _____ Date: _____

2023 Annual Report-Crane Dairy

Final Audit Report 2024-05-20

Created:	2024-05-20
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAENBChL9KXDJiFHEi337SmjW6nowC-0p3

"2023 Annual Report-Crane Dairy" History

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